NAVIER–STOKES / WP19 / REVIEW NOTE

What the v0.29.1 package actually establishes

A corrected, reproducible audit of finite-cutoff records, Python arithmetic, and the Lean verification boundary.

LOCAL PYTHON PASS · 7 tests in each of three modes	**LEAN COMPILER** GitHub Actions result pending
FINITE-CUTOFF RECORDS N = 11 through 18 cross-checked	**CONTINUUM REGULARITY** OPEN · no theorem claimed

VERDICT

This release is a finite-data consistency audit. It is not a proof of global regularity for the three-dimensional Navier–Stokes equations, does not establish a limit as the cutoff tends to infinity, and does not solve the Millennium Problem.

I corrected the package so its labels match what its code can verify: supplied JSON arithmetic is replayed; upstream Arb calculations are not reproduced; unkeyed SHA-256 values are checksums, not signatures.

Prince Upadhyay · Independent Research · 1 November 2026
Repository: github.com/reggaesharkk/navier-stokes-bridge-audit

01 / Review findings

The source report made stronger claims than the included evidence supported. This correction narrows the language and adds checks that reviewers can reproduce.

Area	Correction
Claim language	Removed wording that presented finite-cutoff data as a core mathematical guarantee or implied a path to arbitrary-cutoff regularity. The package now says the continuum question remains open.
Cutoff coverage	The replay now loads and checks each supplied N=11,...,18 JSON record, verifies its metadata and sign fields, and compares every row against the aggregate summary.
Endpoint arithmetic	Displayed endpoints are checked against nominal ± error with a stated 0.0001 tolerance because the supplied values are rounded. This is a consistency check, not a proof of outward rounding.
N14 to N15 comparison	The reported interval is explicitly a difference of supplied signed-numerator endpoints. It is not a difference of normalized G values.
Artifact integrity	Removed the public-constant pseudo-HMAC and its “sealed” status. The manifest now reports unauthenticated SHA-256 checksums only; a checksum does not identify who created a file.
Lean scope	The Lean source now states that its twelve elementary theorems are conditional arithmetic facts. It does not encode the fluid equations or verify the external numerical certificates.

Important boundary: The source package does not contain the raw inputs and trajectories needed to reproduce the upstream Arb computations. The replay validates arithmetic over supplied finite-cutoff records; it does not independently certify those records.

02 | Results and what they mean

The Python code recomputes exact or high-precision arithmetic over the data shipped in the package. The output numbers below should be read with that evidence boundary in mind.

REPLAYED OUTPUT	VALUE	INTERPRETATION
Localized Taylor remainder / prior Q+B	**0.000540356815**	Reported N15 finite-record ratio; not a continuum estimate.
Pairwise perturbation / P	**0.388343411550**	Below one for the supplied N15 arithmetic record.
Algebraic reduction factor	**2016**	Combinatorial factor reported by the finite calculation.
N14 to N15 signed-numerator delta interval	**[1.348e9, 1.017e10]**	Difference of supplied endpoint records; not normalized G.

Verification ladder

1 SUPPLIED RECORDS	2 PYTHON REPLAY	3 LEAN CI	4 PDE CLAIM
JSON files and summary rows	Local arithmetic + 7 tests x 3 modes	Lake build + axiom / nanoda checks	Global regularity remains open

Do not read “PASS” as a Navier–Stokes theorem. It means the specified finite-data checks and regression suite completed successfully. The numerical provenance, rigorous enclosure status, all-cutoff argument, and continuum passage require independent mathematical evidence.

The endpoint display check allows 1e-4 rounding. The code does not verify outward rounding for the original Arb outputs, and the package does not include the original Arb inputs, interval logs, or trajectories.

M14/M15 status: a separate, still-open bridge

M14: v0.29 replays backward steps 239, 238 and 237 using imported strain and residual bounds. This is conditional recurrence arithmetic; a full adjoint-path enclosure is not supplied.

M15: the e219 record is a specification and the joint-pairing pilot is STOP. No modewise M15-versus-M14 error tube or certified full-path signed pairing is supplied.

The N14/N15 interval above does not close this bridge. I could not reproduce the source report's M14-to-M18 transition table without its raw Arb inputs and trajectories.

03 / Software verification

Local Windows run | 1 November 2026

On Windows ARM64 with Python 3.12, I reran the source scanner, the closure replay, and the regression suite. The seven tests passed in normal mode, with `-O`, and with `-OO` (21 test executions total).

Check	Local result	What that result means
Lean declaration scanner	**PASS · 12 declarations**	Text scan only; it does not compile Lean.
Finite arithmetic replay	**PASS**	Recomputed supplied-record consistency checks.
Python tests	**7/7 x normal, -O, -OO**	Regression checks; not a mathematical proof.
Local Lean build	**NOT RUN**	Dependency fetch failed because this environment could not resolve github.com.
GitHub Actions	**Pending at report drafting**	A real Lake build, axiom audit and nanoda job are configured for the PR.

Earlier local ZIP checks

The PowerShell-created archive had 27 manifest-listed files that matched, plus six unlisted Python bytecode-cache files and no saved test logs. That does not make PowerShell unsuitable; it makes that archive less self-documenting. The Ubuntu command shown during review stopped at `unzip: command not found`, so that attempt did not run the tests or create a fresh Ubuntu archive. GitHub Actions is intended to provide a clean Linux run of this corrected source.

CI status will be updated after the GitHub run completes. The source report and release documentation will not describe a Lean build as passing until the workflow actually reports success.

04 / Reproduction and open work

Run from the root of `releases/wp19-v0.29.1-audit-correction/`:

```
python -B tools/check_ns_lean_axioms.py
python -B src/navier_stokes_v029_closure_suite.py
python -B -m unittest discover -s tests -v
python -B -O -m unittest discover -s tests -v
python -B -OO -m unittest discover -s tests -v
sha256sum -c SHA256SUMS.txt

lake update
lake exe cache get
lake build
```

Evidence still needed for stronger claims

- Include and independently inspect the original Arb input files, interval enclosures, and trajectory data; rerun the computations from those inputs.
- Establish that all displayed intervals are outward-rounded rigorous enclosures, with explicit precision and rounding proofs.
- Prove the needed estimates uniformly over every cutoff and justify passage to the continuum.
- Obtain independent mathematical review of the derivations and ensure the Lean formalization encodes the substantive PDE and numerical propositions, if formal proof is intended.

Bottom line: The corrected v0.29.1 package is more honest and easier to review. Its concrete result is a reproducible finite-record audit and a better verification path. The 3D global regularity problem is not solved by this work.

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Source reviewed: Navier-Stokes Bridge Audit Framework Executive Technical Report.pdf (user-provided copy).

M14 / M15 / CURRENT BRIDGE STATUS

A radius is now reported. The signed transfer still does not close.

A related public draft adds an M15 forced-radius replay and then tests the weighted remainder. The replay supplies a conditional scalar L2 radius; the reported remainder budget still fails to preserve the signed margin. This is finite-dimensional audit evidence, not a Navier-Stokes theorem.

LANE	REPORTED RESULT	WHAT IT MEANS
M14 coverage	Three backward steps: 239, 238, 237	The v0.29 replay uses imported segment bounds. It does not enclose the full M14 adjoint path.
M15 forced radius	10.559636070809868141657764813075	The draft reports this over 240 half-steps / 12 shards, conditional on frozen gradient and residual majorants. It is a scalar L2 radius, not a modewise error vector or objective bound.
Weighted remainder	NO CLOSURE	Upper 164,063,087,678,950.09 versus signed lower 5,758,574,435.61; the resulting margin is negative.
Semantic gate	STOP: pilot not authorized	A modewise error vector, componentwise endpoint compatibility, and localized Hessian bound are still missing.

Where the reported budget goes

The defect term is 240,063.83; the support-corrected quadratic remainder is 114,517,700,895,378.76; and the terminal boundary product is 49,545,386,543,507.50. The last 20 half-steps account for about 69.6% of the quadratic upper bound.

Concrete next gate
Certify a modewise error vector on an adjoint-relevant low-mode block, certify the terminal gradient mismatch componentwise, then bound the localized Hessian remainder tightly enough to preserve the signed margin. Keep the tail pilot stopped until those inputs exist. Repeating the scalar radius replay or increasing the cutoff alone does not address the dominant terms.

The M15 forced-radius PASS is conditional on the frozen majorants. The NO_CLOSURE result says the current conservative bounds do not prove the sign; it does not show that the finite-dimensional transfer is false. Neither result implies an all-cutoff estimate or continuum regularity.

SOURCE AND REPRODUCIBILITY

Draft PR #155 (open, unmerged)

Duality audit summary | Semantic-repair gate | Tail STOP record

Source is the public draft branch at commit 35bc96454b471bf37732c7ab216ee199055d80ac. I reviewed its saved records but did not independently rerun its replay programs in this v0.29.1 package.

M14 / M15 FOLLOW-UP · DRAFT PR #155 HEAD 35BC964

The targeted cut helps the budget; it still misses the sign.

I checked the newer M14/M15 draft context against the earlier v0.29.1 replay. These are separate finite-dimensional records. A reported improvement to one slice does not turn the remaining negative margin into a closure.

TRACK	LATEST RECORD	WHAT IT ESTABLISHES
M14	This package: 3 scalar recurrence steps. PR #155: 240-step quadrature on saved reconstructions.	The first uses imported segment bounds; the second is not a certified pairing along the true M15 error path. Neither is a full-path objective transfer.
M15 radius	10.559636070809868141657764813075	Conditional scalar L2 radius over 240 half-steps. It is not a modewise error vector or objective-transfer bound.
Targeted tail	Board v0.6 reports a 0.533% reduction on steps 220–239 and a 0.259% reduction for the full remainder.	Reported in prose only: its claimed output JSON is absent from the PR file list, and the PR says no long Actions run was started. I did not reproduce the numbers.
Sign budget	Remainder 163,638,394,179,339.98 Signed lower bound 5,758,574,435.61 Margin -163,632,635,604,904.37	Even if the reported targeted recomputation is correct, the conservative bound remains far above the signed lower bound. The sign is still open.

Why the next pilot is stopped

The committed active-support gate is **FAIL_ACTIVE_SUPPORT_ROUTE_BLOCKED; NO_VECTOR_PROPAGATION**. The scalar radius is about 6.1 million times the approximate active-Hessian radius threshold, and the fixed aggregate gradient-mismatch bound already exceeds the signed lower bound. The next proof inputs are a certified modewise error vector, componentwise terminal-gradient compatibility, and a useful localized Hessian bound.

A separate earlier M14 export

I found an older v0.27c1 M14 VJP archive on my local drive. Its metadata reports a 241-node binary64 reconstruction, an Arb-versus-FFT check at one interior point, and a 12-mode direct-sum cross-check. The same record explicitly says it contains no residual, trajectory, or quadrature enclosure. I verified the ZIP and its three-entry SHA-256 manifest, but did not recompute the Arb values. It does not supply the missing full-path M15 error vector.

Source and claim boundary

Draft PR #155 — open and unmerged · Attack board v0.6 (source of reported tail improvement)

Active-support feasibility gate · Semantic-repair gate

The attack-board improvement is reported but not independently reproduced here. All statements are limited to finite-dimensional frozen records and the stated conditional arithmetic. Nothing here proves all-cutoff regularity, blow-up exclusion, or continuum Navier–Stokes regularity.